Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/25/2014
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling of Particle Transport in the Melt and Its Interaction with the Solid-Liquid Interface		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	derby@umn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	612-625-8881
Organization Name:	University of Minnesota		
PI Address 1:	Department of Chemical Engineering and Materials Science		
PI Address 2:	151 Amundson Hall, 421 Washington Avenue, SE		
PI Web Page:			
City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	11/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date is 11/30/2015, per NSSC information (Ed., 6/3/15) NOTE: End date is 8/31/2015, per NSSC information (Ed., 6/3/14) NOTE: End date is 8/31/2014, per NSSC information (Ed., 5/30/2013)		
Key Personnel Changes/Previous PI:	Jeffrey Derby is the PI of this NASA project. Collaborators funded by the DLR ParSiWal project are Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth).		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary goal of this research project is to understand the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide and silicon nitride particles during the solidification of silicon. This understanding is vital to improving the quality of solar-grade, multi-crystalline silicon grown by directional solidification processes. This work will coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth). Specifically, finite-element, moving-boundary analyses are being developed and applied to assess the dynamics of particle pushing and engulfment. Of particular interest is to ascertain critical velocities for engulfment for a variety of different particle sizes and thermal gradients. We will also study the effects of shear and solidification flows, in the melt near the interface, on engulfment mechanisms and the effect of a grain boundary at the interface that locally modifies its shape and energy. Planned experiments by ParSiWal in the laboratory as well as during the TEXUS 50 mission are aimed to probe the effects of particle engulfment during silicon solidification and will be used to assess theoretical predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the dynamics of particle pushing and engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production.
Task Progress:	The initial model development is complete, with notable additions to our code to allow for computation of van der Waals forces, pre-melting phenomena, and the Gibbs-Thomson effect of solidification interfacial curvature. We have also completed a set of initial computations involving parametric studies of particle engulfment phenomena. We have found that the model predicts significant departures from prior, idealized analytical analyses under certain conditions. A paper is being prepared to report our progress in a suitable, peer-reviewed journal. Work is ongoing to address the experimental findings of the German ParSiWal project (see the research meeting listed below), and we are preparing a set of computations that will be specifically focused on silicon carbide particles in solidifying silicon. Further model development is also underway to more accurately represent van der Waals forces in the model formulation. The graduate student supported by this project, Mr. Yutao Tao, has completed a set of initial computations and is writing a paper for a peer-reviewed journal. Tao presented a poster of his research at the 6th International Workshop on Crystal Growth Technology, Berlin, Germany, June 15-19, 2014. Derby presented an oral talk on the research results at E-MRS Spring Meeting, Lille, France, May 26-30, 2014. Details on both of these presentations are listed below. Finally, Derby and Tao met on June 20, 2014, with German ParSiWal researchers at the University of Freiburg, Prof. Dr. Arne Croell and his students, to discuss their experiments and future work together. Y. Tao, A. Yeckel, and J.J. Derby, “Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells,” Session W: Crystals for energy conversion and storage, E-MRS Spring Meeting, Lille, France, May 26-30, 2014. Y. Tao, A. Yeckel, and J.J. Derby, “Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells,” Poster presentation, 6th International Workshop on Crystal Growth Technology, Berlin, Germany, June 15-19, 2014.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells." Poster presentation, 6th International Workshop on Crystal Growth Technology, Berlin, Germany, June 15-19, 2014. 6th International Workshop on Crystal Growth Technology, Berlin, Germany, June 15-19, 2014. Abstract book . , Jun-2014
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment during the Growth of Multicrystalline Silicon for Solar Cells." Session W: Crystals for energy conversion and storage, E-MRS (European Materials Research Society) Spring Meeting, Lille, France, May 26-30, 2014. E-MRS (European Materials Research Society) Spring Meeting, Lille, France, May 26-30, 2014. , May-2014